E/P = 2, 2.5, 3: SiPM and ASIC clipping pilot

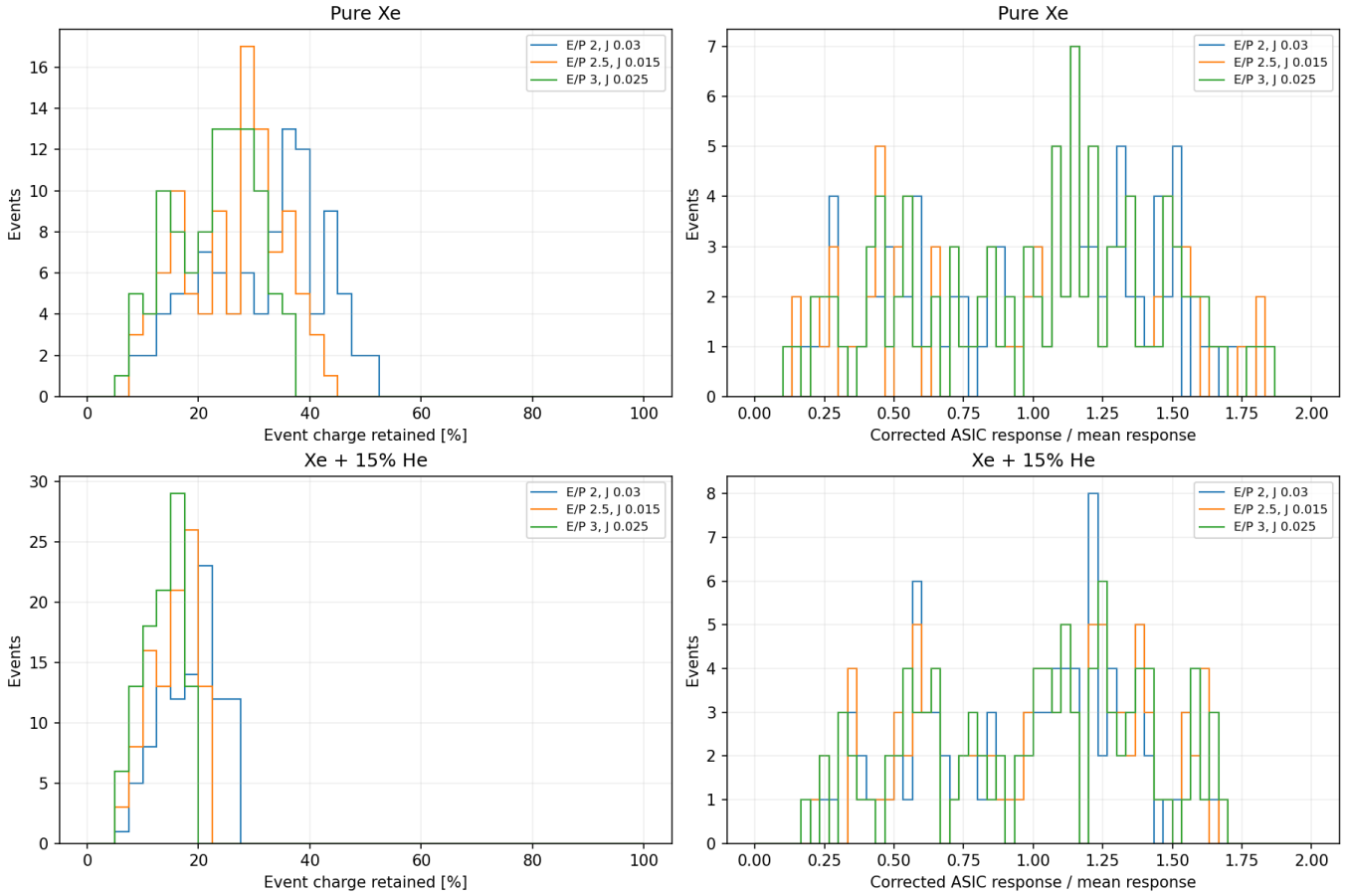
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The same 100 bb0nu source events, primary-pair checksums and drift distances are compared in both gases at E/P = 2, 2.5 and 3 kV/cm/bar. The user-specified EL variance ratios are J = 0.030, 0.015 and 0.025, respectively. The statistical SiPM uses 128 subdivisions per 250 ns bucket, PDE 0.3, gain 2.55e+06 and recovery time 73.4 ns. Electronics remain fixed at 0.5 PE RMS noise and a 1 mA upper current limit (611.9 PE per bucket). Full production remains stopped.

Gas	E/P	J	Events	Charge retained %	Photon cells clipped %	Events with clipping %	Before ASIC RMS %	After ASIC RMS %
Pure Xe	2	0.03	100	32.6	42.2	100	13.6	42.1
Pure Xe	2.5	0.015	100	27.3	49.1	100	16.0	43.4
Pure Xe	3	0.025	100	23.9	54.8	100	17.8	44.2
Xe + 15% He	2	0.03	100	19.3	54.5	100	14.2	37.6
Xe + 15% He	2.5	0.015	100	16.1	60.1	100	16.5	38.6
Xe + 15% He	3	0.025	100	14.2	65.2	100	18.1	39.3

Charge retained is the sum of ASIC output divided by the sum of SiPM input charge over the sample. The clipped-cell percentage uses only SiPM–bucket cells containing incident photons. The event percentage counts events with at least one clipped bucket. These three quantities have distinct denominators.

Event-total response spread is RMS/mean after subtracting the expected stationary dark baseline, correcting known attachment and dividing by deposited energy. SiPM recovery nonlinearity is retained in both columns. The before/after comparison measures the forward response spread under these corrections. All 100 events enter; the quoted widths are full-distribution RMS values. Each plotted ASIC distribution is divided by its own mean.



Numerical results: `results/sipm_statistical_production_pilot/clipping_comparison.h5`. The HTML counterpart links the complete response manifests for each operating point.